

Welcome to one of the most important sections of this course.

New to automation concepts? If terms like APIs, webhooks, or LLMs still feel unfamiliar, we have a beginner-friendly theory section that covers all the fundamentals. You'll find it near the end of the course here: [\(Optional\) Beginner Theory and Foundations - Fundamental Concepts](#)

By now you've built some basic workflows in n8n and you're getting comfortable with the platform. This section will give you the essential knowledge you need to understand and build intelligent AI systems.

We'll start by **exploring vector databases** - the memory systems that make AI agents truly smart. You'll learn what they are, how they work, and most importantly, how to choose the right one for your specific projects.

Next, we'll dive deep into **RAG - Retrieval-Augmented Generation**. This is the technology that allows AI to access external knowledge and give more accurate, up-to-date responses. It's absolutely crucial for building professional AI agents.

Finally, we'll clarify **the differences between automation, AI automation, AI agents, AI assistants, and multi-agent systems**. These distinctions might seem subtle, but understanding them is key to building the right solution for each use case.

This foundational knowledge will prepare you for the advanced AI agent projects we'll build later in the course.

Even if you've worked with AI before, the way we explain these concepts in the context of n8n will give you new insights and practical understanding.

See you in the next tutorial. :)